

Layer-wise Update Frequency for Continual Diffusion Under Nested Learning Paradigm

Names: Amit Sason, Eldar Mamedov and Or Shraga
E-mails: {amitsaso,eldarmam,shragao}@post.bgu.ac.il

Abstract

This project addresses the challenge of "**Catastrophic Forgetting**" in generative diffusion models when trained on sequential tasks. Our objective is to demonstrate a Proof of Concept (POC) for a frequency-based continual learning strategy in DDPMs that outperforms existing regularization methods in retaining prior knowledge while successfully acquiring new tasks. Addressing this challenge is critical for the deployment of adaptable AI systems, as standard neural networks suffer from **catastrophic forgetting** - the rapid degradation of performance on previously learned tasks upon exposure to new data.

Taking inspiration from the human brain's multi-time-scale update mechanisms (Nested Learning - NL), we propose a training strategy where different layers of a U-Net architecture are updated at varying frequencies. Specifically, shallow layers are updated frequently, while deeper layers integrate information over longer, slower cycles. We evaluate our approach using a class conditional Denoising Diffusion Probabilistic Model (DDPM) on a partitioned MNIST dataset: Task A (digits 0-5) followed by Task B (digits 6-9). Our results demonstrate that this simple architectural tweak significantly reduces forgetting. The model demonstrates superior retention of past tasks, **improving FID A from 22.3 to 8.5**. The Nested Learning model maintained higher image quality for previous tasks and has better FID and utility forgetting scores proving that multi-frequency updates provide a robust pathway for continual generative learning.

Relevant Papers

1. **Behrouz, A., et al. (2025).** *Nested Learning: The Illusion of Deep Learning Architectures*. Google Research.
2. **Ho, J., et al. (2020).** *Denoising Diffusion Probabilistic Models*. NeurIPS 2020.
3. **Wang, Z., et al. (2025).** *Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models*. Georgia Institute of Technology, arXiv preprint.

Problem Description

Diffusion probabilistic models (DPMs) have become a dominant approach for high fidelity image generation due to stable training and excellent sample quality. However, most diffusion training assumes a stationary dataset. In practical settings, data often arrives sequentially: new classes, new domains, or updated distributions. When a model is fine-tuned on new data, performance on earlier data frequently degrades, a phenomenon known as **catastrophic forgetting**.

This challenge is particularly pervasive in the lifecycle of Large Language Models (LLMs), where continuous training on fresh data (e.g., recent news or new coding libraries) frequently causes 'knowledge erosion', leading models to hallucinate or fail on facts and reasoning patterns they had previously mastered.

Background and Related Work

Diffusion Probabilistic Models: Denoising Diffusion Probabilistic Models (DDPMs) learn to reverse a fixed noising process that gradually corrupts data into Gaussian noise. Training is typically performed by predicting either the noise added at a given timestep or the denoised sample, using a U-Net backbone conditioned on timestep embeddings (and optionally class labels). DDPMs are known for stable likelihood-based training and strong perceptual quality.

Continual Learning and Catastrophic Forgetting: Continual learning aims to learn tasks sequentially while maintaining performance on earlier tasks. Naïve fine-tuning often leads to catastrophic forgetting because parameters that encode earlier tasks are overwritten. Common mitigation strategies include regularization-based methods (e.g., EWC), replay-based approaches, parameter isolation, and architectural methods.

Nested Learning (NL) and Multi-Timescale Updates: The Nested Learning perspective argues that layered systems (biological or artificial) can benefit when **different layers adapt at different rates**. Intuitively, lower-level features (edges/strokes) should remain stable, while higher-level representations adapt more quickly. Translating this concept to deep neural networks suggests controlling update frequencies (or step sizes) across modules to preserve general features and reduce interference when learning new tasks.

Continual Learning in Generative Models: Continual learning for generators is challenging because forgetting affects both fidelity and coverage. For diffusion models, this is especially important: forgetting may show up as missing modes (older classes disappear), degraded sample quality for old tasks, or increased confusion between classes. Therefore, evaluation should measure both *perceptual distance* (e.g., FID) and *semantic correctness* (e.g., classifier accuracy on generated images).

Chosen Method for Solving

This project investigates catastrophic forgetting in diffusion models using a controlled continual learning benchmark: **Split-MNIST**. We train a class-conditional DDPM in two stages. We compare:

1. **Baseline sequential diffusion training ($A \rightarrow B$)**: standard training continuation.
2. **Nested-Learning-inspired diffusion training ($A \rightarrow B$)**: a modification where different parts of the network are updated at different frequencies, motivated by the NL hypothesis that hierarchical systems benefit from multiple timescales of adaptation.
3. **Elastic Weight Consolidation (EWC) training ($A \rightarrow B$)**: a regularization-based approach that constrains the model during Task B training by adding a penalty term to the loss function, which discourages significant changes to parameters identified as critical for Task A via the Fisher Information Matrix.

The key question is: Can NL-style frequency separation reduce forgetting in diffusion models without significantly harming generation quality on new classes?

Datasets and Preprocessing

We use **MNIST**: Standard benchmark (60k train, 10k test) (1x28x28 grayscale handwritten digits) with a training set of 60,000 examples, and a test set of 10,000 examples.

- **Dataset Sub-setting:**
MNIST was categorized into:
Task A: Digits [0, 1, 2, 3, 4, 5].
Task B: Digits [6, 7, 8, 9].
- **Normalization:** Images were normalized to a range between [0, 1].
- **Task Partitioning:** Task A consists of 60k training samples (subset), and Task B consists of the remaining digits to mimic learning a large task followed by a smaller one.

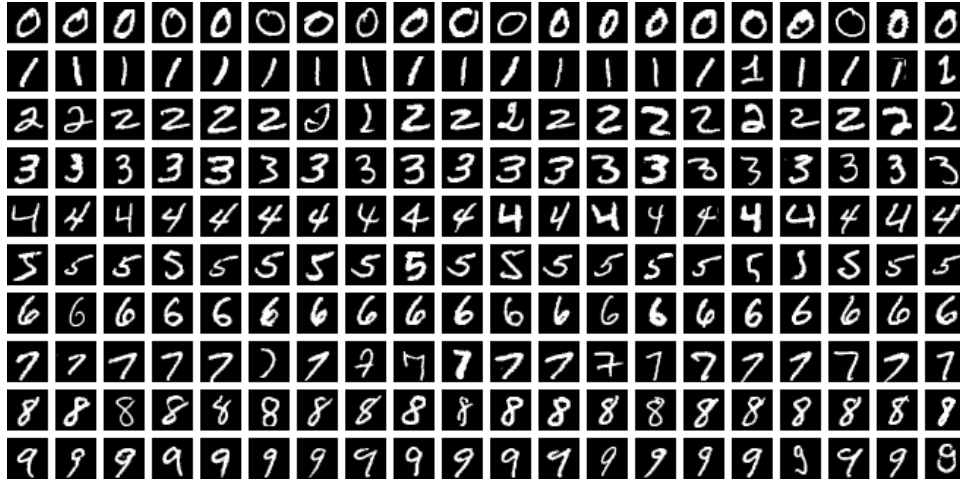


Figure 1 – Mnist dataset- examples of digits from 60k images.

Architecture

The proposed architecture utilizes a **U-Net** architecture for DDPM (**Context-Unet**) backbone partitioned into three logical depth levels, each governed by a specific update frequency to mitigate catastrophic forgetting.

The **Shallow** level, comprising the `init_conv`, `up2`, and `out_head` modules, operates at full resolution and is updated at every optimization step ($k = 1$) to capture fine-grained visual details. The **Middle** level, which includes `down1`, `up1`, and their corresponding embedding layers (`time_embed2`, `context_embed2`), processes intermediate features and is updated every 4 batches ($k = 4$).

The **Deep** level, responsible for the most abstract semantic representations at the bottleneck, consists of `down2`, `to_vec`, `up0`, and the primary embedding layers (`time_embed1`, `context_embed1`), with updates restricted to every 16 batches ($k = 16$).

Conditioning information is injected via learnable class embeddings (representing the digit labels), which are projected to a dimension of 128 channels and concatenated with the feature maps at the Middle and Deep levels to guide the generative process.

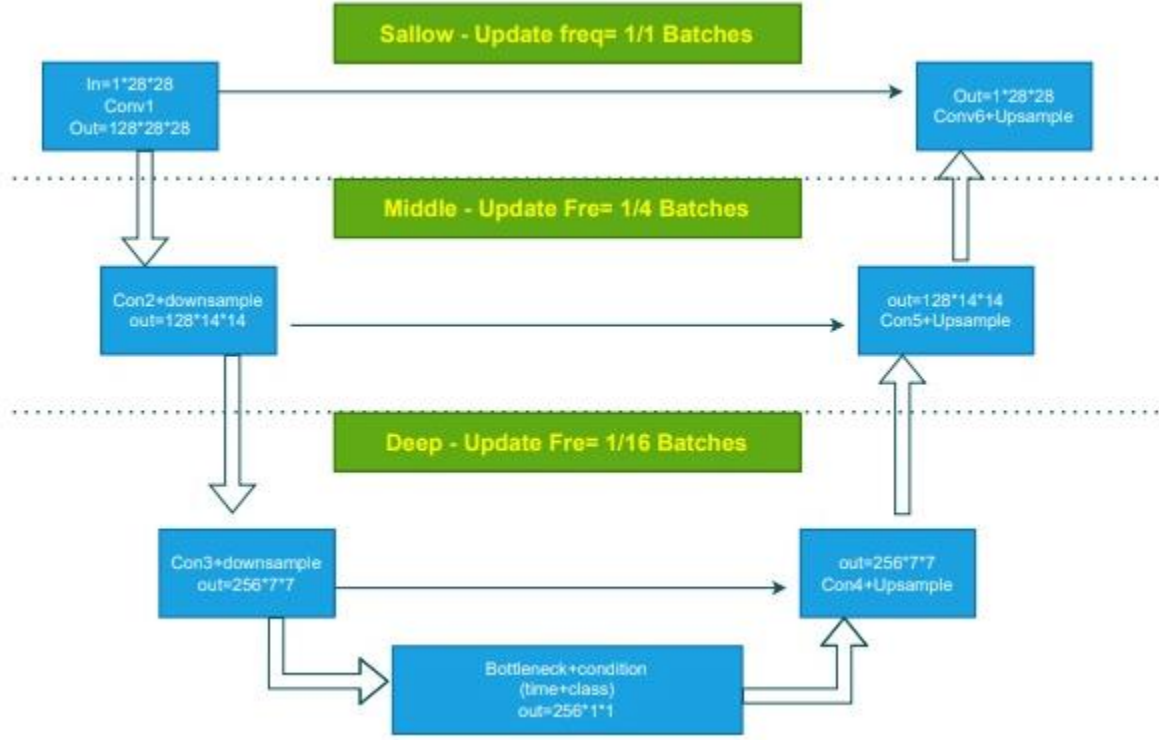


Figure 2 – Unet architecture partitioned into three logical depth levels.

Loss Function

To train the diffusion model, we employ the standard denoising objective characteristic of Denoising Diffusion Probabilistic Models (DDPMs). Specifically, the network is optimized to minimize the Mean Squared Error (MSE) between the actual Gaussian noise, ϵ added to the image at time-step 't' and the noise predicted by the model, ϵ_θ where the model estimates the noise added at time step 't' given class condition 'c'. The loss function is defined as:

$$(1) \quad L = \|\epsilon - \epsilon_\theta(x_t, t, c)\|^2$$

Crucially, we do not incorporate any auxiliary regularization terms typically associated with continual learning (such as Elastic Weight Consolidation or replay buffers). By relying solely on this naive objective, we isolate the impact of our layer-wise update frequency schedule, demonstrating that the mitigation of catastrophic forgetting arises strictly from the Nested Learning-inspired optimization dynamics rather than explicit loss constraints.

Inference

The inference process focuses on synthesizing new samples from the trained model to evaluate its generative capability and retention of past tasks. The procedure follows the standard reverse diffusion trajectory:

- **Initialization:** The process begins by sampling a latent tensor, x_T , from a standard Gaussian distribution, $x_T \sim N(0, I)$. This pure noise tensor serves as the starting point for generation.
- **Reverse Diffusion Process:** The model performs a sequence of $T = 400$ iterative denoising steps. At each step 't' (from 400 down to 0), the network predicts the noise component ϵ_θ and subtracts it from the current state, gradually refining the chaotic noise into a structured 28X28 grayscale image.
- **Class Conditioning:** Generation is guided by specific digit labels (e.g., forcing the model to generate a "3"). As observed in the implementation, we utilize a guidance scale (typically $\omega=2.0$) to sharpen the distinction between classes and improve image fidelity.
- **Evaluation Protocol:** To ensure statistically significant metrics, we generated **1,000 synthetic images for each digit class**. These samples were then used to calculate the Fréchet Inception Distance (FID) scores, allowing us to quantify the quality of the generated images and the severity of catastrophic forgetting.

Training and experiments

Throughout the project, we investigated the proposed method by applying different update frequencies to each layer, comparing its performance against both a standard baseline and the EWC method.

Baseline: Class-Conditional DDPM

Our baseline is a class-conditional DDPM implemented with a U-Net architecture that takes as input:

- Noisy image x_t
- Timestep t embedding,
- Class label $y \in \{0, \dots, 9\}$ embedding

The forward process adds Gaussian noise over $T=400$ steps:

$$(2) \quad q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

The model learns the reverse denoising process by predicting noise ϵ . The training objective is:

$$(3) \quad \mathcal{L} = \mathbb{E}_{x_0, t} [\| \epsilon - \epsilon_{\theta}(x_t, t, y) \|_2^2]$$

Sequential training is done by continuing optimization from the Task-A checkpoint while training only on Task-B samples.

Proposed: NL-Inspired Frequency-Based Updates

We implement a NL-inspired modification that introduces different update frequencies across network components (or groups of layers). The main idea is to reduce how often (or how strongly) certain parts of the model are updated during training, to better preserve learned semantic knowledge of Task-A while training on Task-B.

We define 3 groups: shallow, middle, deep, suggesting that layers closer to the input data should be updated more frequently than deep layers containing more semantic features. Each group is updated according to a frequency parameter, for a “slow/deep” group, gradients are accumulated and applied only every several batches. For a “fast/shallow” group, gradients are applied every batch.

We explored a set of update schedules S based on geometric progressions, specifically powers of 2. Defining the schedule as $S = \{[f_{\text{shallow}}, f_{\text{middle}}, f_{\text{deep}}]\}$, our experimental search space included: $S = [1, 2, 8], [1, 4, 8], [1, 4, 16], [2, 4, 8], [2, 16, 32]$. Empirical evaluation determined that the schedule $[1, 4, 16]$ provided the optimal performance. This aligns with the hypothesis that deep layers require significantly lower update frequencies to preserve semantic knowledge.

Comparison Model: Elastic Weight Consolidation (EWC)

To benchmark our frequency-based approach, we compare it against Elastic Weight Consolidation (EWC), a widely established regularization method for continual learning. EWC mitigates catastrophic forgetting by estimating the importance of each network parameter for the previous task (Task-A) using the diagonal of the Fisher Information Matrix. During the training of Task-B, a quadratic penalty term is added to the standard diffusion loss, which constrains the “important” parameters to remain close to their optimal values from Task-A while allowing less critical parameters to adapt to the new data. We control the rigidity of this constraint using a regularization coefficient (λ).

Two-Stage Training Protocol

Both baseline and NL models follow the same protocol:

1. Training proceeds sequentially: train on Task A to convergence by early stop training loss of 0.025.
2. Save model checkpoint as Model-A.
3. Initialize from Model-A and train on Task B (digits 6–9) until convergence / early stopping.
4. Save final checkpoint as Model A→B.
5. Evaluate Task-A for both Model A and Model A→B, quantify the catastrophic forgetting for Task-A by the agreed metrics.

Convergence was set to 100 Epochs or Early stopping condition defined as reaching Empirical Loss of 0.025. Optimizer type was set to ADAM, 3 optimizers were set, one for every depth group.

```
n_epoch = 100
batch_size = 256
n_T = 400 # number of diffusion steps
device = ("cuda:0" if torch.cuda.is_available() else "cpu")
n_classes = 10
n_feat = 128
lrate = 1e-4
Drop_prob = 0.1
early_stop_loss = 0.025
update_frequency = [1,4,16] #[shallow,middle,deep]
```

Novelty of method with respect to literature

Unlike standard regularization-based methods (such as EWC) or replay-based approaches that rely on modifying the loss function or maintaining external memory buffers, our method introduces novelty by fundamentally altering the optimization trajectory itself. We implement the Nested Learning (NL) paradigm proposed by Behrouz et al. (2025), which views the neural network not as a monolith, but as a Continuum Memory System (CMS) comprised of nested components operating at different time scales.

The primary advancement of our work is the application of this paradigm specifically to Denoising Diffusion Probabilistic Models (DDPMs). While the original NL research largely focused on language or sequence models (e.g., Titans, HOPE), we provide one of the first empirical explorations of layer-wise update frequencies in a generative computer vision context. By integrating a multi-time-scale update schedule directly into the U-Net training loop where shallow layers update frequently (high plasticity) and deep layers update sparsely (high stability). We mimic biological learning processes to "freeze" core semantic knowledge during new task acquisition without explicit architectural expansion or regularization penalties.

Definition of Success and Training Goals

Success is defined by the model's ability to maintain high performance on Task A after being trained on Task B without seeing Task A data again.

- **Quantitative Metrics:**
 - **Accuracy:** > 90% for Task A digits after Task B training.
 - **FID (Fréchet Inception Distance):** Lower increase in FID for Task A after Task B training (lower is better).
 - **FID Forgetting:** Target a reduction of at least 50% compared to the naive baseline.
- **Qualitative:** Visual clarity of digits 0-5 in the final model.

Evaluation

We evaluate:

1. **Retention:** Does the final model (after B training) still generate digits from Task A with good quality and diversity?
2. **Plasticity:** Does the model learn Task B effectively (high quality for digits 6–9)?
3. **Overall sample quality:** How does sequential training affect global metrics like FID and Accuracy?

Fréchet Inception Distance (FID) To evaluate the perceptual quality and diversity of the generated images, we employ the FID metric, adapted specifically for the MNIST domain. Instead of using the final classification output (logits), which only represents class probabilities, we extract high-dimensional feature embeddings from the **penultimate layer** of a pre-trained CNN classifier. This layer captures rich structural information—such as stroke width, curvature, and geometric patterns—providing a deeper representation of the image distribution. We then calculate the mean and covariance of these features for both the real and generated datasets, computing the Fréchet distance between their multivariate Gaussian distributions. A lower FID score indicates that the generated images are statistically like real data in terms of both quality and variation.

Classification Accuracy (The "Judge") To assess the semantic consistency of the generated samples, we utilize a pre-trained MNIST classifier that serves as an impartial "Judge." For every image generated by our model, we feed it into this classifier to obtain a predicted label. We then compare this prediction to the intended target label (e.g., verifying that a generated "3" is indeed recognized as a "3"). This metric is crucial for quantifying **Catastrophic Forgetting**: by monitoring the accuracy on the first task (Task A) after training on the second task (Task B), we can directly measure the extent to which the model has lost its ability to produce recognizable digits from the previous distribution.

1. Base Measurements

- **Original FID A (fid_A_prev):**

Measures the generation quality of Task A (digits 0-5) immediately after training on Task A. This serves as the **baseline performance** before any catastrophic forgetting could occur.

- **Current FID B (fid_B_current):**

Evaluates how well the model learned the new task (digits 6-9) by comparing the generated samples to the real MNIST test set for Task B after the sequential training is completed.

2. Retention Measurements

- **Current FID A (fid_A_current):**

Assess the quality of Task A images (0-5) generated by the model **after** it has been updated to learn Task B. A high score here compared to the baseline indicates significant degradation in image quality.

3. Derived Metrics

- **Average FID (AFID):**

Calculated as the arithmetic means of the final FID scores for both tasks.

$$(4) \quad AFID = \frac{FID_{A_{current}} + FID_{B_{current}}}{2}$$

This metric provides a holistic view of the model's overall performance across all learned tasks.

- **FID Forgetting (fid_forgetting_A):**

Quantifies the **degradation in visual quality** for the first task. It is calculated by subtracting the baseline FID from the current $FID_{A_{current}} - FID_{A_{prev}}$ where a higher positive value represents severe forgetting.

- **Accuracy Forgetting (acc_forgetting_A):**

Measures the loss in **semantic recognizability**. It is defined as the drop in classification accuracy for Task A digits from the end of the first phase to the end of the second phase $ACC_{A_{prev}} - ACC_{A_{current}}$.

Results

Examples

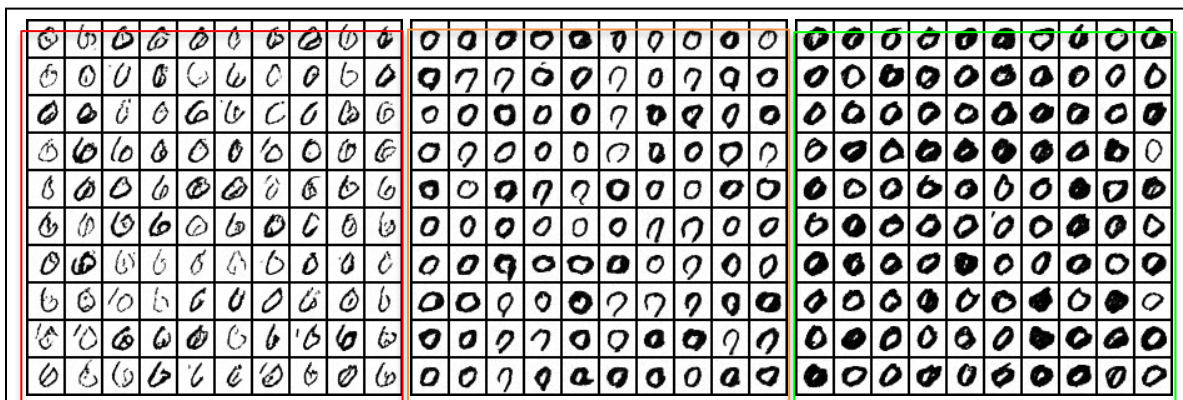


Figure 3 - Digit 0 task B after A, baseline(left) vs EWC (middle) vs frequencies 1 4 16 (right).



Figure 4 - Digit 2 task B after A, baseline(left) vs EWC (middle) vs frequencies 1 4 16 (right).

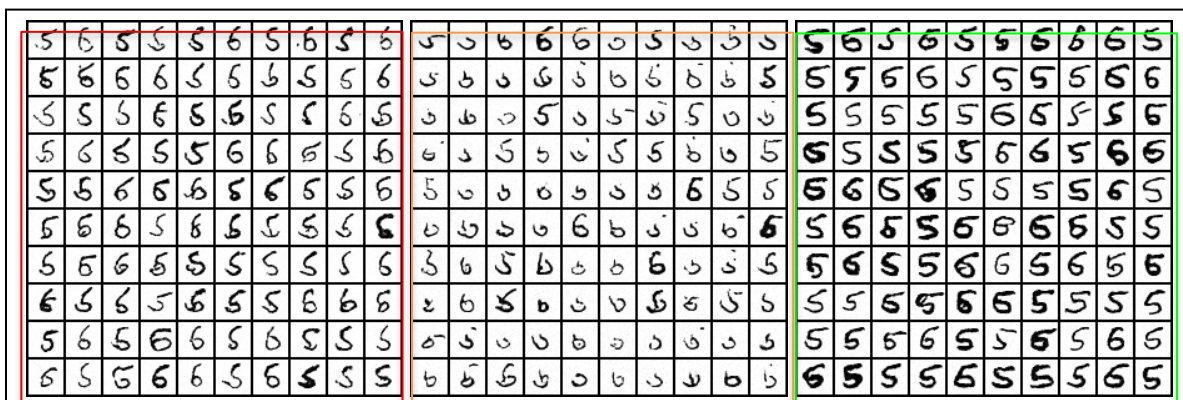


Figure 5 - Digit 5 task B after A, baseline(left) vs EWC (middle) vs frequencies 1 4 16 (right).

The above generated examples of digits from task A after leaning task B. From the three examples above, it is evident that the NL model generates Task A digits that closely resemble the target. In contrast, the other two models output ambiguous hybrids or undefined shapes.

Confusion matrix

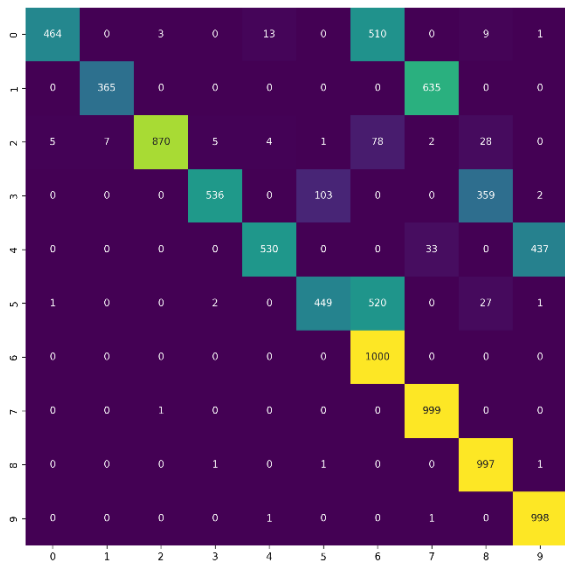


Figure 6- Confusion matrix baseline

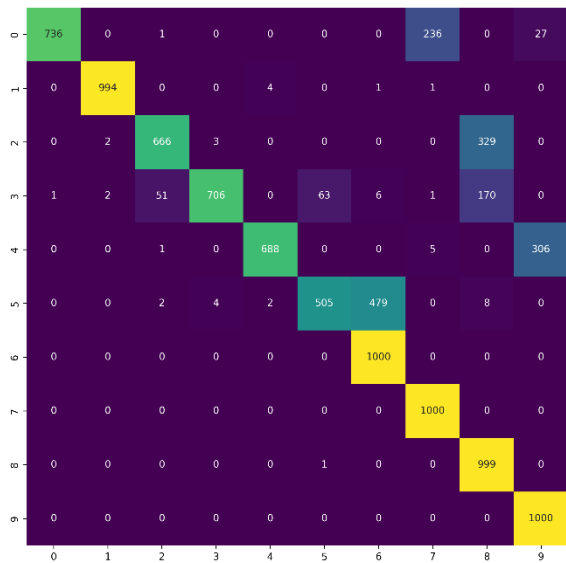


Figure 7- Confusion matrix EWC

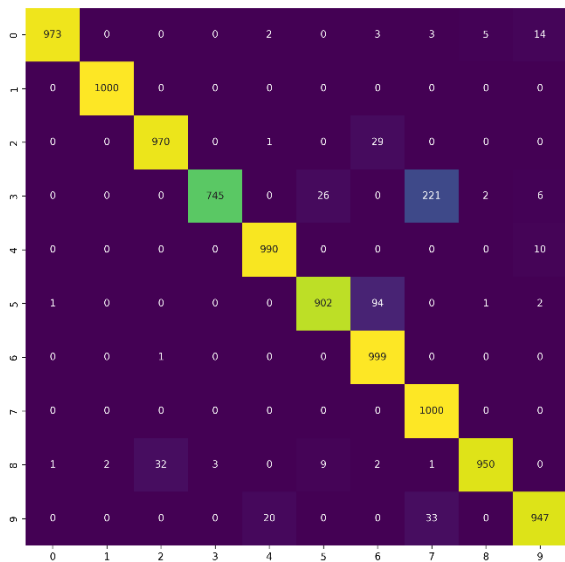


Figure 8- Confusion matrix frequency base 1 4 16

Analysis of the confusion matrix reveals a clear performance hierarchy for Task A. The Baseline model performs poorly, while the EWC (Elastic Weight Consolidation) model shows moderate improvement. The Frequency-based approach achieves the highest accuracy for Task A; however, this gain comes at a cost. Unlike the Baseline and EWC models which maintain 100% accuracy across Task B the frequency-based model experiences a 5% performance degradation in the final two digits of Task B.

FID and accuracy tables

Table 1 FID and accuracy:

Model Name	baseline	EWC	freq 1_2_8	Freq 1_4_8	freq 1_4_16	freq 2_4_8	freq 2_16_32
Acc A prev	1	1	1	0.999	1	0.999	1
Accuracy A	0.536	0.716	0.627	0.743	0.93	0.855	0.765
Accuracy B	0.998	1	0.999	0.999	0.974	0.996	0.997
Accuracy AB	0.721	0.829	0.776	0.845	0.948	0.911	0.858
FID A Prev	12.957	9.305	15.677	19.422	17.905	15.143	18.27
FID A	49.491	45.425	58.678	44.323	26.492	33.185	41.826
FID B	13.755	18.512	18.512	19.799	15.097	19.936	20.899
FID AB avg	31.623	31.968	38.595	32.061	20.794	26.56	31.363

Results for the model and version tested in the project. Freq X_Y_Z related to as $S = \{[f_{\text{shallow}}, f_{\text{middle}}, f_{\text{deep}}]\}$, "ACC A prev" and "FID A prev" are original results taken after trained on task A only. "Accuracy A", "Accuracy B", "Accuracy AB", " FID A", " FID B", "FID AB" result tested after additional train for task B.

Comparative Results. While the NL [1,4,16] model showed slightly higher initial FID on Task A ('FID A prev'), it demonstrated remarkable robustness against catastrophic forgetting. Post-training maintained a Task A FID of **26.492**, whereas Baseline and EWC degraded to **49.491** and **45.425**, respectively. Learning capability for the new task ('FID B') remained consistent across all models.

Table 2 - compares loss of baseline, EWC to our best frequency-based model:

Model Name	baseline	EWC	freq 1_4_16
A Accuracy loss	0.464	0.284	0.07
A FID loss	36.534	36.12	8.587

Summary

This project investigated the challenge of catastrophic forgetting in generative models by evaluating three distinct approaches: a standard Baseline, Elastic Weight Consolidation (EWC), and a Frequency-based strategy. The objective was to maintain the performance of Task A (initial digit generation) after the model transitioned to learning Task B.

Our results demonstrate a clear performance:

- **The Baseline model** suffered significantly, achieving only **53.6% accuracy** for Task A.
- **EWC** provided moderate preservation with **71.6% accuracy**.
- **The Frequency-based approach (specifically 1_4_16)** emerged as the superior method for retention, achieving a high of **93% accuracy** for Task A and the best overall Fréchet Inception Distance (FID) of **20.794**.

These findings suggest that frequency-based regularization is highly effective at preserving early-task knowledge in generative pipelines.

Additional remarks that you'd like us to be aware of

While the frequency-based approach provided the best preservation of Task A, several critical observations were noted during the experiment:

- **Performance Trade-offs:** The gain in Task A accuracy via the frequency-based model came at a slight cost to Task B. Specifically, the model experienced a **5% performance degradation** in the final two digits of Task B, whereas the Baseline and EWC models maintained near-perfect accuracy (100%) for the second task.
- **Visual Fidelity:** Analysis of the generated images and FID scores shows that the Frequency 1_4_16 configuration significantly reduced visual artifacts compared to the Baseline, which showed clear signs of "forgetting" through distorted Task A outputs.

- **Future Work:** Further research could focus on optimizing the frequency base parameters to eliminate the slight degradation in Task B, potentially exploring adaptive frequency scaling to balance retention and new learning more dynamically.